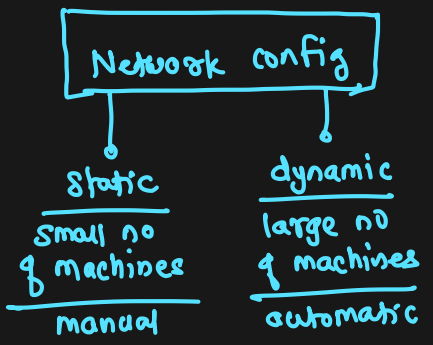
DHCP Server

dynamically assigns IP address

DCHP Server

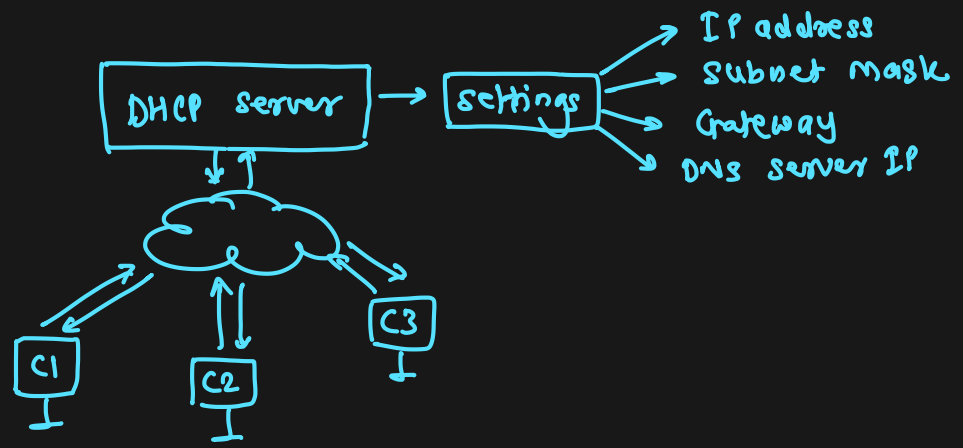
- Subnet mask
- Gateway
- DNS server

- DHCP is a network management protocol used on IP networks
- Its primary function is to **automatically assign IP addresses** and other **network configuration parameters** to devices (hosts) so they can communicate on the network
- Without DHCP, network administrators would have to manually configure IP addresses, subnet masks, gateways, and DNS servers for every device, which is impractical for large networks (like Wi-Fi hotspots or corporate networks)
- **Key Characteristics**
  - **Protocol Type:** Client-Server model
  - **Transport Layer:** Uses **UDP** (User Datagram Protocol)
  - **Ports:** Server listens on **UDP Port 67**; Client listens on **UDP Port 68**
  - **Broadcast Domain:** Initial communication uses broadcasts because the client doesn't have an IP address yet
  - **Standards:** Defined in RFC 2131 (IPv4) and RFC 3315 (DHCPv6)
- **Ports**
  - DHCP Server: UDP port 67
  - DHCP Client: UDP port 68



Why DHCP ?

- **Automation:** Reduces administrative overhead
- **Avoids Conflicts:** Prevents two devices from getting the same IP address
- **Mobility:** Devices get valid IPs automatically when moving between networks
- **Scalability:** Essential for large networks where devices join and leave frequently (BYOD, IoT)
- **Centralized Management:** Administrators can change network settings (like DNS servers) in one place, and all devices get the update
- **IP Address Reuse:** When a device leaves the network, its IP address is returned to the pool for reuse, preventing exhaustion of addresses



Key Components

- **DHCP Server** : program (service | daemon)
  - A device (router or server) that holds IP addresses and configuration.
- **DHCP Client** : program (ipconfig)
  - Any device requesting an IP (laptop, phone, printer).
- **Scope** 192.168.1.150 to 192.168.1.160 → address pool (collection of addresses)
  - A range of IP addresses available for lease to clients.
- **Lease** : when lease time is over, the client renews settings
  - The time duration for which an IP is assigned to a client.
- **Reservation** : useful for servers
  - A permanent lease (static mapping) of an IP to a specific MAC address.
- **Options**
  - Additional parameters (e.g., default gateway, DNS server, domain name).

DHCP Process

- **Step 1: DHCP Discover (Client -> Server)**
  - The client boots up without an IP
  - It broadcasts a DHCPDISCOVER message to the entire network (Destination: 255.255.255.255)
  - Source IP: 0.0.0.0, Destination Port: 67
  - Goal: "Is there a DHCP server out there?"
- **Step 2: DHCP Offer (Server -> Client)**
  - All DHCP servers that receive the Discover message respond with a DHCPOFFER
  - The offer contains a proposed IP address, subnet mask, lease duration, and the server's IP
  - If multiple servers respond, the client picks one.
  - Goal: "I have an IP for you."
- **Step 3: DHCP Request (Client -> Server)**
  - The client broadcasts a DHCPREQUEST message
  - It requests the specific IP offered by the chosen server
  - This broadcast also informs other DHCP servers that their offers were declined
  - Goal: "I accept your offer."
- **Step 4: DHCP Acknowledgment (Server -> Client)**
  - The chosen server sends a DHCPACK (Acknowledgment)
  - This contains the final configuration (IP, DNS, Gateway) and starts the lease timer
  - If the IP is no longer available (e.g., taken by someone else in the meantime), the server sends a DHCPNAK (Negative Acknowledgment), and the client must restart the process
  - Goal: "Here is the IP and settings; you are good to go."

DHCP Lease and Renewal Process

- An IP address is not given forever; it is "leased."
- **Lease Time:** The duration the IP is valid (e.g., 24 hours, 7 days)
- **T1 Timer (50% of Lease):** When 50% of the lease time has passed, the client sends a **unicast DHCPREQUEST** to the server that gave it the IP to renew the lease
- **T2 Timer (87.5% of Lease):** If the server doesn't respond to the T1 renewal, the client waits until 87.5% of the lease has passed. It then **broadcasts** a DHCPREQUEST to any available DHCP server to renew or get a new IP
- **Expiry (100%):** If no response is received by the end of the lease, the client must stop using the IP and restart the DORA process

DHCP Messages

- DHCPDISCOVER
  - Client to Server (broadcast)
  - Client searches for a DHCP server
- DHCPOFFER
  - Server to Client (unicast/broadcast)
  - Server offers an available IP address
- DHCPREQUEST
  - Client to Server (broadcast)
  - Client accepts the offer
- DHCPACK
  - Server to Client (unicast)
  - Server confirms assignment and provides lease + options
- DHCPNAK
  - Server to Client
  - Denies the request (e.g., IP no longer available or invalid)
- DHCPDECLINE
  - Client to Server
  - Client detects IP is already in use (e.g., via ARP)
- DHCPRELEASE
  - Client to Server
  - Client voluntarily gives up its IP address
- DHCPINFORM
  - Client to Server
  - Client already has an IP (manually) but needs other options (DNS, gateway)

DHCP Allocation Methods

- **Dynamic Allocation**
  - The server assigns an IP from a pool for a limited time (lease)
  - Most common method
  - Used for general clients (laptops, phones)
- **Automatic Allocation** → deprecated
  - The server assigns a permanent IP to a client from the pool
  - Once assigned, the IP is tied to that client's MAC address forever
  - Rarely used today
- **Manual Allocation (Static DHCP / Reservations)**
  - The administrator assigns a specific IP to a specific MAC address
  - The client always gets the same IP
  - Used for servers, printers, and routers

Best Practices



- **Only one DHCP server per network** - Disable DHCP on your router to avoid conflicts
- **Server needs static IP** - DHCP servers should not obtain their own IP via DHCP
- **Plan your IP range** - Ensure your DHCP pool doesn't overlap with static IPs used by servers, printers, or network equipment
- **Backup configuration** - Always backup /etc/dhcp/dhcpd.conf before making changes
- **Monitor lease exhaustion** - Use tools like dhcpd-pools to monitor IP usage